

Build a Langchain Streamlit Chatbot for EDA using LLMs

Project Overview

Overview

Data analysis and visualization play a pivotal role in modern business operations, providing invaluable insights that drive strategic decision-making and organizational success. For business stakeholders, the ability to interpret and derive meaning from vast datasets is crucial in understanding market trends, customer behavior, and operational performance. Data analysis enables stakeholders to identify patterns, correlations, and outliers within datasets, uncovering hidden opportunities and potential risks. Visualization complements this process by transforming complex data into intuitive charts, graphs, and dashboards, facilitating clear and concise communication of insights.

Large Language Models (LLMs) present a paradigm shift in how non-technical stakeholders interact with data. These powerful natural language processing models enable users to converse with data in plain language, bypassing the need for specialized technical skills or assistance from data experts. By leveraging LLMs, non-technical stakeholders can directly query data, receive relevant insights, and make informed decisions in real-time, without relying on intermediary teams.

This project involves the development of a Streamlit chatbot integrated with Lanchain and OpenAI to facilitate conversational interactions with a SQL database. Leveraging the Langchain framework, the chatbot interprets natural language instructions provided by the user, generating corresponding SQL queries to extract relevant data from the database. Through seamless integration with data visualization libraries such as Matplotlib or Plotly, the chatbot dynamically generates Python code to create visualizations based on the retrieved data, offering insightful insights and analytics in response to user queries.

Prerequisite Project: Kindly ensure the completion of the [LLM Project for building and fine-tuning a large language model](#) before proceeding with this project.

Note: Please note that the use of the OpenAI API may require the utilization of allocated free credits or additional purchases (depending on the account status) for implementing the project. Kindly take note of the free credits limit provided for your usage.

Aim

The aim of the project is to develop a Streamlit chatbot integrated with Langchain technology, enabling natural language interactions with a SQL database to generate visualizations and provide insights in real-time. Through this, the project seeks to streamline data exploration and analysis processes, making them more intuitive and accessible to users.

Data Description

The e-commerce dataset consists of 7 interconnected datasets covering distribution centers, user events, inventory items, order details, products, and user profiles.

- Distribution Centers dataset: Contains information about distribution centers including unique identifiers, names, and geographical coordinates.
- Events dataset: Provides details about user events including unique identifiers, timestamps, user IDs, session IDs, and event types.
- Inventory Items dataset: Includes data on inventory items such as unique identifiers, timestamps for creation and sale, product details, and distribution center associations.
- Order Items dataset: Contains information on ordered items including unique identifiers, timestamps for creation, shipping, delivery, and return, along with associated product and user IDs.
- Orders dataset: Provides details about orders including unique identifiers, timestamps for creation, shipping, delivery, and return, along with order status and quantity of items.
- Products dataset: Includes information about products such as unique identifiers, costs, retail prices, categories, brands, and distribution center associations.
- Users dataset: Contains user profiles with details like unique identifiers, names, email addresses, demographics, location coordinates, traffic sources, and account creation timestamps.

Tech Stack

→ Language: Python 3.10.4

- Libraries: mysql-connector-python, mysqlclient, pydantic, langchain, matplotlib, plotly, streamlit
- Model: Open AI GPT-4

Approach

- Database Creation:
 - Install the MYSQL and MYSQL workbench on your system
 - Creating a new MYSQL database and loading the CSVs in to the database
- Prompt Engineering Strategies:
 - Acting as a Role
 - Language Translation
 - Travelling Guide
 - Combining LLMs and Prompts in multi-step workflows
- Langchain Integration:
 - Creating SQL agent that transform natural language queries into SQL queries
 - Creating Python script agent that creates visualizations from SQL query outputs
- Streamlit Chatbot Creation
- Integrating Memory and Guardrails

Cost Breakdown

Components

- Streamlit Chatbot with OpenAI and experimentation in OpenAI

Cost Breakdown as of January 17, 2025

The cost might have been updated, it is recommended to go through the following pricing pages to come up with updated guessestimates.

Open AI Pricing <https://openai.com/api/pricing/>

Component	Description	Usage	Estimated Cost

API Calls	OpenAI API (GPT-4 Turbo and GPT-3.5 Turbo) for chatbot responses	Models used: - GPT-3.5 Turbo (16K context): \$0.0015 per 1K tokens (input), \$0.002 per 1K tokens (output)- GPT-4 Turbo (8K context): \$0.01 per 1K tokens (input), \$0.03 per 1K tokens (output)	Total Estimated Usage:- 100 queries- Avg 1,000 tokens/query Cost Estimate:- GPT-3.5 Turbo: (input: \$0.15, output: \$0.20)- GPT-4 Turbo: (input: \$1.00, output: \$3.00)
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Total Estimated Cost: ~\$5 - \$10 per project.

Recommendation: GPT-3.5 Turbo is more cost-effective for high-volume interactions, while GPT-4 Turbo provides superior responses but at a higher cost. Proper token usage monitoring is essential to optimize costs.

Experimentation should balance performance needs with budget constraints. For cost efficiency, consider using GPT-3.5 Turbo for routine chatbot interactions and GPT-4 Turbo for complex or critical tasks requiring higher accuracy.

Modular code overview:

Once you unzip the code_folder.zip file, you can find the following:

```
├─ data
│   ├── distribution_centers.csv
│   ├── ecommerce-data-description.docx
│   ├── ecommerce.db
│   ├── events.csv
│   ├── inventory_items.csv
│   ├── orders.csv
│   ├── order_items.csv
│   ├── products.csv
│   └── users.csv
├─ Query_To_Dashboard.ipynb
├─ Readme.md
├─ requirements.txt
└─ src
    ├── agent.py
    ├── chat_app.py
    ├── helper.py
    ├── stream_lit_demo.py
    └── test
```

Here is a brief information on the files:

- The code folder contains all the Python scripts and sample files used in the project.
- The README.MD and the solution methodology file provides essential information about the project, usage, and setup instructions.
Kindly follow all the instructions for running the code from Readme.md file
- The requirements.txt file has all the required libraries with respective versions.
Kindly install the file by using the command **pip install -r requirements.txt**

Project Takeaways

1. Understand LLM fundamentals, attention mechanism, and transformer architecture
2. Learn about various applications of LLMs in different industries
3. Introduction to Langchain and its functionalities
4. What is Prompt Engineering?
5. Implement prompt engineering scenarios for communication with LLMs
6. Learn to set up a MYSQL database
7. Understand the complexities involved in generating queries and codes using LLMs
8. Create Langchain agents for SQL query generation from natural language
9. Create Langchain agents for Python code generation from natural language
10. Gain insight into memory and guardrails, essential concepts in managing and optimizing the performance of LLMs
11. Development of a Streamlit chatbot interface for intuitive data interaction and visualization